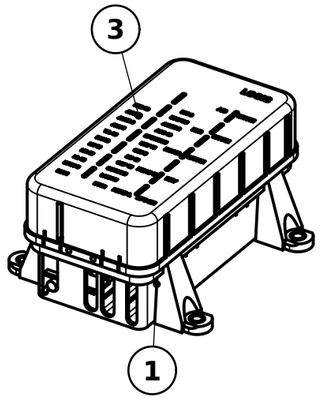
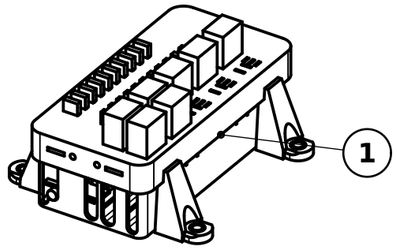
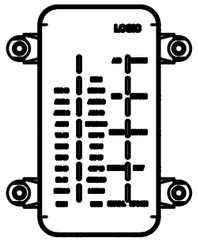
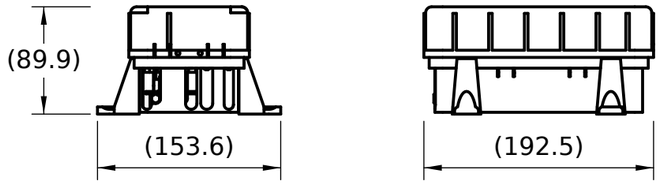



**TRIMETRIC VIEW**




**TRIMETRIC - COVER REMOVED**






**FRONT / TOP / RIGHT VIEWS  SCALE 1:5**

**REVISIONS**

| REV | DESCRIPTION | BY | DATE |
|---|---|---|---|
| P | #273: FB-105 base M, FB-106 pan F, FB-107 cover H, FB-115 / -116 strips D | Claude | 16/10/26 |
| R | #275: FB-105 base N | Claude | 17/10/26 |
| S | #297: FB-105 base P, FB-106 pan G, FB-107 cover J, FB-115 / -116 strips E | Claude | 18/10/26 |
| T | #303: FB-105 base R, FB-106 pan H, FB-107 cover K | Claude | 18/10/26 |
| U | #308: FB-105 base S; first assembly drawing (buildability review 18/10/26) | Claude | 18/10/26 |
| V | FB-105 base T (strip hold-downs) | Claude | 18/10/26 |

| ITEM | QTY | PART NUMBER | DESCRIPTION | MATERIAL / SPEC |
|---|---|---|---|---|
| 1 | 1 | ST185-FB-105 | LOGIC BASE | PA12, 3D PRINTED |
| 2 | 1 | ST185-FB-106 | LOGIC PAN (BELLY PAN) | PA12, 3D PRINTED |
| 3 | 1 | ST185-FB-107 | LOGIC COVER | PA12, 3D PRINTED |
| 4 | 1 | ST185-FB-115/-116 | LOGIC BUS STRIPS BUS1 + BUS2 | C110 Cu, TINNED |
| 5 | 19 | PEM S-M4-2ZI | SELF-CLINCHING NUT M4, PRESSED INTO THE STRIP | C110 Cu, TINNED |
| 6 | 2 | PEM SS-M5-2ZI | SELF-CLINCHING NUT M5, PRESSED INTO THE STRIP | C110 Cu, TINNED |
| 7 | 4 | ATM-2 | MINI BLADE FUSE 2 A (ATM) | PURCHASED |
| 8 | 3 | ATM-5 | MINI BLADE FUSE 5 A (ATM) | PURCHASED |
| 9 | 4 | ATM-7-1/2 | MINI BLADE FUSE 7.5 A (ATM) | PURCHASED |
| 10 | 2 | ATM-10 | MINI BLADE FUSE 10 A (ATM) | PURCHASED |
| 11 | 7 | ATM-15 | MINI BLADE FUSE 15 A (ATM) | PURCHASED |
| 12 | 1 | ATM-20 | MINI BLADE FUSE 20 A (ATM) | PURCHASED |
| 13 | 1 | 4-1904124-2 | MICRO RELAY (ALL 7 POSITIONS) | PURCHASED |
| 14 | 1 | 4-1904124-3 | MICRO RELAY: HORN (SEE NOTE 2) (NOT SHOWN) | PURCHASED |
| 15 | 2 | 5-1393292-8 | MICRO RELAY: HEAD LH, HEAD RH (SEE NOTE 2) (NOT SHOWN) | PURCHASED |
| 16 | 3 | 6-1419137-4 | MICRO RELAY: K_ACC, K_EPS_EN, K_REV (SEE NOTE 2) (NOT SHOWN) | PURCHASED |
| 17 | 14 | 280919-2 | MICRO RELAY COIL RECEPTACLE 4.8 X 0.8 | PURCHASED |
| 18 | 17 | 42281-1 | MICRO RELAY LOAD RECEPTACLE 6.3 X 0.8 | PURCHASED |
| 19 | 22 | 1-968849-1 | MINI FUSE RECEPTACLE MCP 2.8, 20 / 18 AWG | PURCHASED |
| 20 | 20 | 1-968851-1 | MINI FUSE RECEPTACLE MCP 2.8, 16 / 14 AWG | PURCHASED |
| 21 | 1 | RNF-100-1/8-0-STK | INSULATION BUILD-UP SLEEVE 1/8 (NOT SHOWN) | PURCHASED |
| 22 | 1 | RNF-100-3/32-0-STK | INSULATION BUILD-UP SLEEVE 3/32 (NOT SHOWN) | PURCHASED |
| 23 | 4 | 94459A140 | M3 HEAT-SET INSERT (OPTIONAL COVER BACKUP) (NOT SHOWN) | BRASS |
| 24 | 4 | SSCF-M3-10-A4 | M3 x 10 CAP SCREW (OPTIONAL COVER BACKUP) (NOT SHOWN) | A4 STAINLESS |
| 25 | 11 | 320554 | INPUT JUMPER RING LUG, 22-16 AWG, #8 / M4 | Cu, TINNED |
| 26 | 8 | 53941-1 | INPUT JUMPER RING LUG, 16-14 AWG, #8 / M (SEE NOTE 2) (NOT SHOWN) | Cu, TINNED |
| 27 | 1 | M22759/16-14-2 | INPUT JUMPER WIRE 14 AWG, ~60 MM EACH | PURCHASED |
| 28 | 1 | M22759/16-16-2 | INPUT JUMPER WIRE 16 AWG, ~60 MM EACH (SEE NOTE 2) (NOT SHOWN) | PURCHASED |
| 29 | 1 | M22759/16-18-2 | INPUT JUMPER WIRE 18 AWG, ~60 MM EACH (SEE NOTE 2) (NOT SHOWN) | PURCHASED |
| 30 | 1 | M22759/16-20-2 | INPUT JUMPER WIRE 20 AWG, ~60 MM EACH (SEE NOTE 2) (NOT SHOWN) | PURCHASED |
| 31 | 19 | RS 287-3915 | HEX BOLT M4 X 8, INTO THE PEM NUT IN THE STRIP | A2 STAINLESS |
| 32 | 19 | RS 231-0691 | SPRING WASHER M4 UNDER EACH JUMPER BOLT HEAD | A2 STAINLESS |
| 33 | 2 | Klauke 2R5 | TUBULAR LUG 10 MM2, M5 HOLE | Cu, TINNED |
| 34 | 2 | RS 287-3921 | HEX BOLT M5 X 10, FEED LUG INTO THE PEM SS-M5… | A2 STAINLESS |
| 35 | 4 | EFC 10S00056 | COMPRESSION LIMITER, FEET (4) | STEEL, ZINC |

## NOTES:

1. OVERALL SIZES ARE FOR REFERENCE.
2. PARTS THAT SHARE ONE MODEL (THE MICRO RELAYS, LUGS, JUMPER WIRES) ARE BALLOONED ON THE FIRST ITEM OF THEIR GROUP.
3. USE THE M5 x 10 FEED BOLT, NOT A SHORTER ONE: ITS TIP GOES 1.7 INTO THE SPINE AND LOCKS THE STRIP.
4. SHEET 2: WHAT EACH FUSE AND RELAY FEEDS. SHEETS 3 AND 4: THE HARNESS WIRE AT EACH TERMINAL, NAMED AS ON THE HARNESS DRAWINGS.
5. TERMINALS GO IN FROM UNDER THE DECK, CONTACT BOX FIRST, UNTIL THE LANCE CLICKS. PULL THE WIRE BACK GENTLY TO CHECK IT HOLDS (TE 114-2025 3.6, 114-20021 3.7).
6. KEEP HARNESS TIES AT LEAST 38 mm BELOW THE DECK, AND DO NOT BEND THE WIRES HARD TO ONE SIDE AT THE DECK (TE 114-20021 3.7 C).

**UNLESS OTHERWISE SPECIFIED:**
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5  X.X ±0.2
X.XX ±0.05  ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
**ST185 CELICA GT-FOUR  -  LINK ECU CONVERSION**

TITLE
# LOGIC FUSE BOX - ASSEMBLY

| PART / DWG NO. | REV | MATERIAL |
|---|---|---|
| ST185-FB-100L | V | SEE BOM |

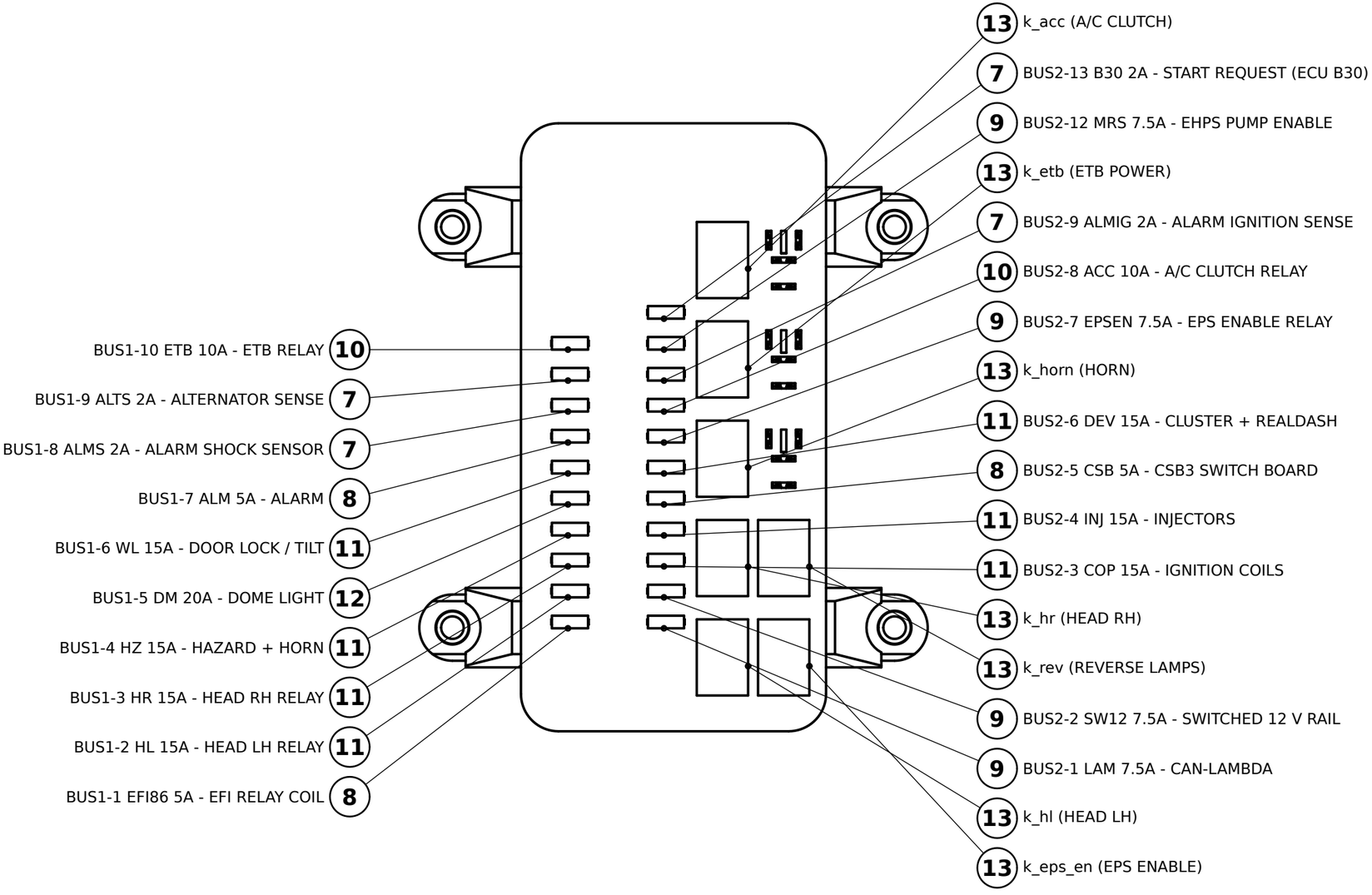
| SCALE | UNITS | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|---|
| 1:5 | mm | B (11x17) | 1 OF 6 | 2026-10-18 | SteveCAD |

SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

# WHAT EACH FUSE AND RELAY FEEDS (COVER OFF)

TOP VIEW  SCALE 1:1.5




**UNLESS OTHERWISE SPECIFIED:**
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
**ST185 CELICA GT-FOUR - LINK ECU CONVERSION**

TITLE
# LOGIC FUSE BOX - ASSEMBLY

| PART / DWG NO. | REV | MATERIAL |
|---|---|---|
| ST185-FB-100L | V | SEE BOM |

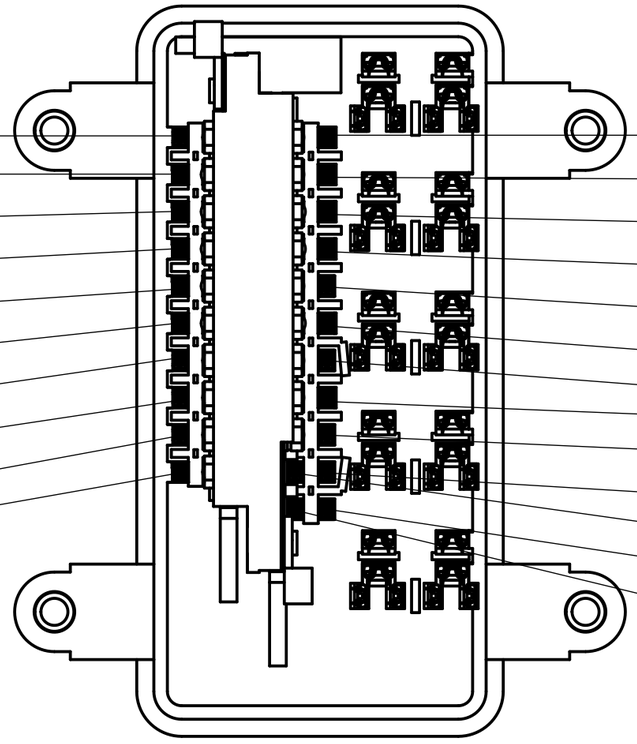
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SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

# HARNESS WIRES AT THE FUSES - SEEN FROM BELOW, BELLY PAN OFF

BOTTOM VIEW  SCALE 1:1.5



BUS1-1 > POWER R2 86 - EFI coil+

BUS1-2 > HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

BUS1-3 > HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

BUS1-4 > HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

BUS1-5 > OEM J/B 1 plug 1H pin 2 (LH kick), L-Y wire - DOME 20A feed (EWD 1993 p.132)

BUS1-6 > OEM old R/B2 relay lead pin 2 (LH kick), W-L wire - door lock + tilt-away (EWD p.137)

BUS1-7 > Alarm +12 V splice (AntiTheft loom, center dash) - alarm pin 1 constant +12 V

BUS1-8 > Shock sensor +12 V splice (AntiTheft loom, ECU area) - shock sensor pin A

BUS1-9 > Alternator plug G2 pin 1 (engine bay, via loom C) - S battery sense (EWD 1993 p.74-75)

BUS1-10 > LOGIC M14 30 - ETB 12V in

BUS2-1 > Bulkhead B pin 95 (DRB102, RH firewall) - CAN-Lambda +12 V

BUS2-2 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 1 - SW12 +12 V bus

BUS2-3 > Bulkhead A pin 2 (DRB102, RH firewall) - COP coil +12 V rail (COP 15A)

BUS2-4 > Bulkhead A pin 1 (DRB102, RH firewall) - injector +12 V rail (INJ 15A)

BUS2-5 > CSB3 plug pin 1 (CSB3 box under the cluster) - CSB3 +12 V switched

BUS2-6 > Dash device +12 V splice (CAN loom) - center cluster and RealDash Pi +12 V

BUS2-7 > LOGIC M21 30 - EPS EN 12V in

BUS2-8 > LOGIC M15 30 - A/C 12V in

BUS2-9 > Alarm ignition splice (AntiTheft loom, center dash) - alarm pin 3 ignition sense

BUS2-12 > Bulkhead A pin 42 (DRB102, RH firewall) - EHPS pump logic enable (pump pin C1)

BUS2-12 IN > LOGIC M21 87 - EPS EN 12V out

BUS2-13 > ECU plug B pin 30 (Link ECU, center dash) - DI 7 start request

BUS2-13 IN > OEM starter relay socket R/B4 pin 1, B-W wire - ignition ST1 start request (EWD 1993 p.66)

**UNLESS OTHERWISE SPECIFIED:**
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X  ±0.5  X.X  ±0.2
X.XX  ±0.05  ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
**ST185 CELICA GT-FOUR  -  LINK ECU CONVERSION**

TITLE
# LOGIC FUSE BOX - ASSEMBLY

| PART / DWG NO. | REV | MATERIAL |
|---|---|---|
| **ST185-FB-100L** | **V** | SEE BOM |

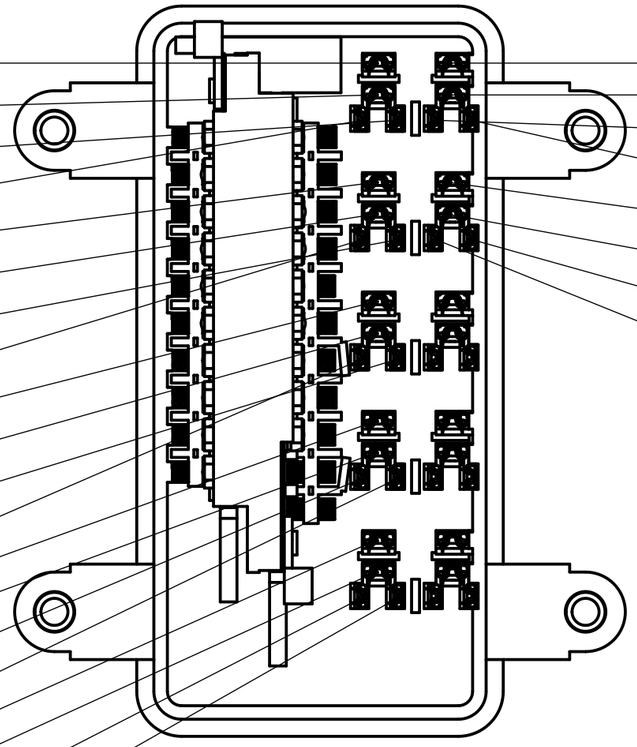
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SOURCE
C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

# HARNESS WIRES AT THE RELAYS - SEEN FROM BELOW, BELLY PAN OFF

BOTTOM VIEW  SCALE 1:1.5



k_hl 30 > HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

k_hl 87 > LH headlight plug H1 pin 3 (loom C, LH front) - headlight LH +12 V (EWD 1993 p.116)

k_hl 86 > HEAD LH splice - HL 15A fuse to HEAD LH relay pins 30 and 86

k_hl 85 > Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2

k_hr 30 > HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

k_hr 87 > RH headlight plug H2 pin 3 (loom C, RH front) - headlight RH +12 V (EWD 1993 p.116)

k_hr 86 > HEAD RH splice - HR 15A fuse to HEAD RH relay pins 30 and 86

k_hr 85 > Headlight coil splice - HEAD LH and RH relay coils to cowl plug IE1 pin 2

k_horn 30 > HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

k_horn 87 > Horn splice (loom C, front) - horn relay output to horns H3 and H4 pin 1

k_horn 86 > HAZ-HORN splice - HZ 15A fuse to the hazard feed and the horn relay

k_horn 85 > OEM cowl plug IE1 pin 19 (LH kick), G wire - horn switch line (EWD 1993 p.185)

k_etb 30 > FB-LOGIC BUS1-10 - ETB 10A

k_etb 87 > ECU plug B pin 5 (Link ECU, center dash) - ETB motor supply from the ETB relay

k_etb 85 > ECU plug A pin 20 (Link ECU, center dash) - Aux 2 sinks the ETB relay coil

k_etb 86 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 5 - SW12 +12 V bus

k_acc 30 > FB-LOGIC BUS2-8 - ACC 10A

k_acc 87 > A/C magnet clutch A4 pin 4 (loom C, compressor) - clutch +12 V (1990 EWD p.152)

k_acc 85 > ECU plug B pin 10 (Link ECU, center dash) - Ign 8 sinks the A/C clutch relay coil

k_acc 86 > A/C pressure switch splice (A-cabin loom) - switch A5 pin 4 output to the clutch relay

k_eps_en 30 > FB-LOGIC BUS2-7 - EPSEN 7.5A

k_eps_en 87 > FB-LOGIC BUS2-12 IN - MRS in

k_eps_en 85 > ECU plug B pin 12 (Link ECU, center dash) - Ign 6 sinks the EPS enable relay coil

k_eps_en 86 > SW12 bus plug DT06-12 (glove box, by LOGIC box) 9 - SW12 +12 V bus

k_rev 30 > GAUGE 15A splice - J/B 3 feed to reverse relay pin 30 and the back-up switch

k_rev 87 > Back-up lamp splice (RearLights loom, rear) - R11 and R12 pin 4 back-up lamps

k_rev 86 > Reverse switch splice (B-cabin loom) - back-up switch B1 pin 1, +12 V in reverse

k_rev 85 > POWER box NEG bar M8 stud (glove box) - reverse relay coil ground

**UNLESS OTHERWISE SPECIFIED:**
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
**ST185 CELICA GT-FOUR - LINK ECU CONVERSION**

TITLE
# LOGIC FUSE BOX - ASSEMBLY

| PART / DWG NO. | REV | MATERIAL |
|---|---|---|
| **ST185-FB-100L** | **V** | SEE BOM |

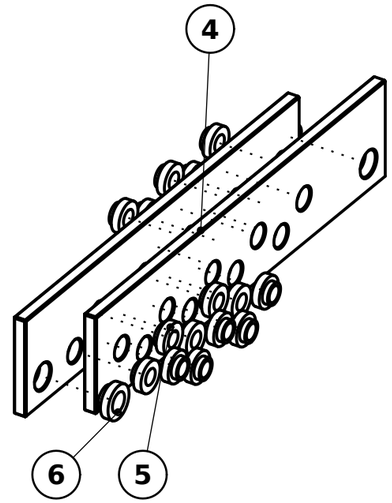
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| 1:1.5 | mm | B (11x17) | 4 OF 6 | 2026-10-18 | SteveCAD |

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3RD ANGLE

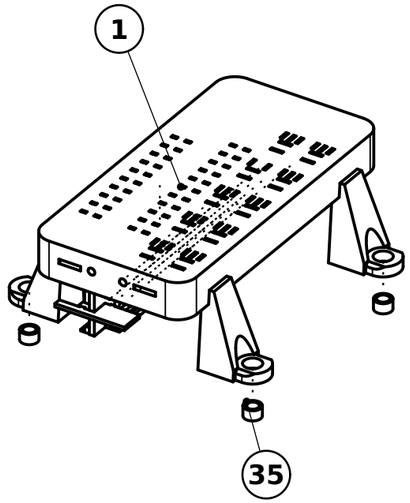
## STEP 1: PRESS A PEM NUT INTO EVERY STRIP HOLE

EXPLODED TRIMETRIC (NOT TO SCALE)



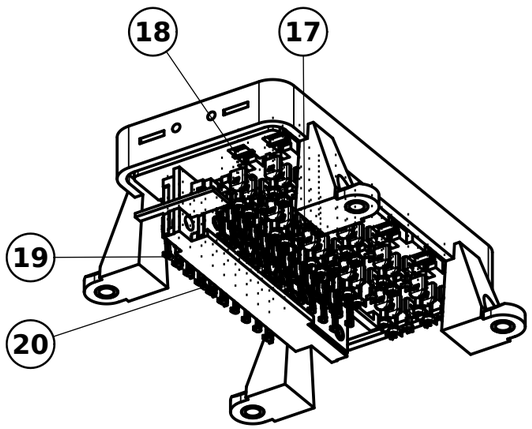

## STEP 2: LIMITERS; STRIPS IN FROM THE REAR TILL THE HOOK CLICKS

EXPLODED TRIMETRIC (NOT TO SCALE)



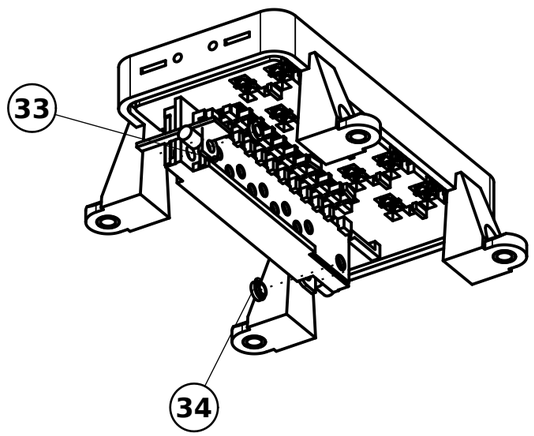

## STEP 3: TERMINALS UP INTO THE CAVITIES

EXPLODED, SEEN FROM BELOW (NOT TO SCALE)




## STEP 4: FEED LUGS ON THE M5 x 10 FEED BOLTS

EXPLODED, SEEN FROM BELOW (NOT TO SCALE)




**UNLESS OTHERWISE SPECIFIED:**
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
**ST185 CELICA GT-FOUR - LINK ECU CONVERSION**

TITLE
# LOGIC FUSE BOX - ASSEMBLY

| PART / DWG NO. | REV | MATERIAL |
|---|---|---|
| **ST185-FB-100L** | **V** | SEE BOM |

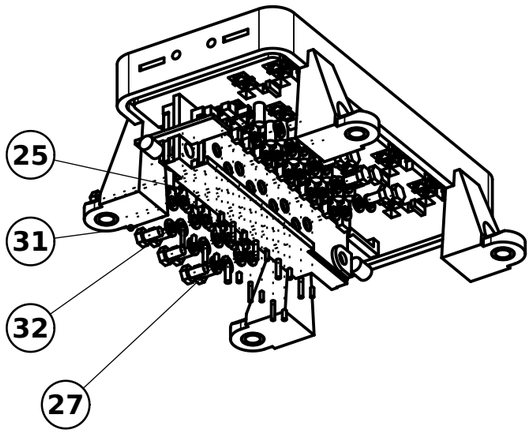
| SCALE | UNITS | SIZE | SHEET | DATE | DRAWN |
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| AS SHOWN | mm | B (11x17) | 5 OF 6 | 2026-10-18 | SteveCAD |

SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE

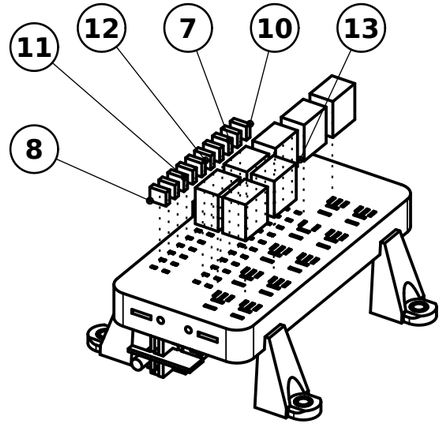
## STEP 5: JUMPERS: WIRE, LUG, WASHER, M4 x 8 BOLT

EXPLODED, SEEN FROM BELOW (NOT TO SCALE)



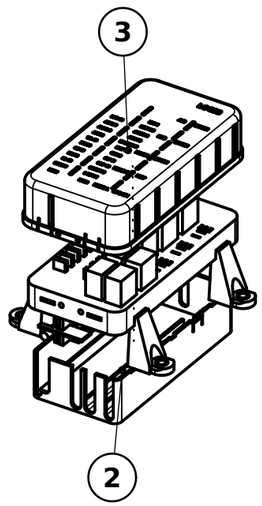

## STEP 6: FUSES AND RELAYS IN FROM THE TOP

EXPLODED TRIMETRIC (NOT TO SCALE)




## STEP 7: BELLY PAN AND COVER

EXPLODED TRIMETRIC (NOT TO SCALE)




**UNLESS OTHERWISE SPECIFIED:**
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
**ST185 CELICA GT-FOUR - LINK ECU CONVERSION**

TITLE
# LOGIC FUSE BOX - ASSEMBLY

| PART / DWG NO. | REV | MATERIAL |
|---|---|---|
| **ST185-FB-100L** | **V** | SEE BOM |

| SCALE | UNITS | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|---|
| AS SHOWN | mm | B (11x17) | 6 OF 6 | 2026-10-18 | SteveCAD |

SOURCE C:\\projects\\active\\st185-link-ecu-config\\cad\\fusebox\\rev7\\ST185_fusebox_rev7_assembly.step

3RD ANGLE